

Department of Mathematics
Assignment-II

Subject: Mathematics-I for ME/CV/EEE stream

Sub Code: BMATM/C/E101

Q.NO	Assignment-II	BLT	CO's
1	Prove that with usual notation $\tan \phi = r \frac{d\theta}{dr}$	L1,L2&L3	CO1
2	Find the pedal equation for the curve $\frac{2a}{r} = 1 + \cos \theta$	L1,L2&L3	CO1
3	Using Maclaurin's expansion Prove that $\sqrt{1 + \sin 2x} = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24}$	L1,L2&L3	CO2
4	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.	L1,L2&L3	CO2
5	Evaluate $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dz dy dx$	L1,L2&L3	CO4
6	Derive the relationship between the beta and gamma functions.	L1,L2&L3	CO4

